Mechanisms and Forecasting of Financial Volatility: Herding, Percolation, and Price–Volume Analysis

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Collective dynamics, inference and market risk
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How does collective coordination enter financial price–volume fluctuations, and how much mechanism information survives aggregation? We study a finite Erdős–Rényi herding model through forward simulation, inverse inference, and empirical compatibility tests. Cluster moments connect coordination to aggregate fluctuations. Population moments identify parameters within the specified model family, while exact observational equivalence limits recovery of internal network edges. Finite-size percolation does not guarantee a pure power law over the selected return range. Joint BTC observations also challenge the fixed activity and observation interface. A separate application uses historical observables to forecast next-day volatility. Across the original 183 held-out dates, the full model reduces log-RV error by 31.20% relative to persistence. Its 2.26% point improvement over HAR has an interval crossing zero. An existing US-equity tool supports omitted-risk diagnosis, strategy-redundancy screening, and allocation comparisons. The physics supplies mechanism relations, conditional inversion, and compatibility restrictions; overall forecasting skill does not establish a causal herding effect. This report connects the research questions to completed analyses, with source figures, numerical tables, code, data-access routes and reproduction instructions.

I. RESEARCH QUESTIONS AND SCOPE

The project addresses three connected questions with distinct evidential requirements. Q1 asks how coordination, network connectivity, and percolation change aggregate price–volume fluctuations. Q2 asks which parameters remain recoverable, which structural information aggregation removes, and whether market observations satisfy necessary model relations. Q3 asks whether historical observables and temporal structure support next-day volatility analysis. Practical usefulness means providing testable, reproducible analysis; outperforming every established method is not a requirement.

Table I connects the research strands to the completed analyses and their evidence.

TABLE I. Research scope and evidence. Literature, model calculations and market observations address complementary questions.

Research strand	Completed work and report location	Evidence scope
Theory and computation; statistical physics	Sections II–V: rules, moments, generating functions, finite-graph inversion, and simulation	Both theory and computation
Complex networks; emergence	Trader graphs, connected clusters, and cluster moments governing aggregate fluctuations; Figs. 1 and 2	Trader interactions differ from asset-correlation networks
Complex adaptive systems	Section I A: opinion/strategy switching, feedback, and an analytically tractable static baseline	Dynamic adaptation covered through literature; not attributed to the independent ER baseline
Models of return power laws	Herding model; Section III and Appendix A: cluster and return distributions	Finite herding model; rejected tail fits retained
Bubbles and crashes	Cont–Bouchaud, Lux–Marchesi, and JLS comparison in Table II	Mechanisms and published cases reviewed; no crash-date forecast
Simulations or real data	1,080-graph scan, independent simulation studies, BTC and US-equity observations	Both types provided and reported separately
Criticality and scaling	Threshold, finite-size transition, cluster distributions, temporal aggregation, and magnitude memory	No universal exponent or real-market critical-point claim

Evidence labels guide interpretation. **MODEL** denotes derivations and exact checks under explicit assumptions; **SIMULATION** denotes numerical experiments with known mechanisms. **REAL DATA** denotes market observations and historical evaluations; **LITERATURE** denotes models and findings from published sources.

A. Model literature, bubbles, and crashes

Cont–Bouchaud links random communication, shared cluster actions, and truncated power laws[1]. We retain the coordination structure while specifying volume observation and unit-order execution. The paper's size and activity limits do not automatically describe our finite experiments. Lux–Marchesi introduces opinion and trader-type switching to generate endogenous broad tails and temporal structure[2]. These provide an analytically tractable baseline and a dynamic-feedback comparison, respectively.

JLS studies reinforcing imitation, conditional bubble growth, and stochastic crash risk[3]. Its simplified setup gives

$$dp/p = \mu(t)\,dt - \kappa\,dj, \quad \mathbb{E}[dj] = h(t)\,dt, \quad \mu(t) = \kappa h(t). \tag{1}$$

The hazard $h(t)$ links growth conditional on survival to compensation for jump risk. The paper discusses pre-crash patterns for Wall Street in 1929, 1962, and 1987, and Hong Kong in 1997. These are retrospective literature findings, not our reproductions or predictions. The bubble endpoint differs from the random crash time; a critical time is not a deterministic crash date.

TABLE II. Three physical models address different explanatory tasks. Findings attributed to these papers are literature evidence.

Model	Mechanism and purpose	Role in this project
Cont–Bouchaud[1]	Shared actions within random groups; aggregate fluctuations and truncated power laws	Finite ER implementation, with price–volume observation, inversion, and tests
Lux–Marchesi[2]	Switching driven by trends, group opinion, and relative profits; adaptation and memory	Literature comparison explaining feedback absent from the independent ER baseline
JLS[3]	Reinforcing imitation, hazard, and conditional bubble growth	Bubble/crash review; no LPPL fitting or crash-date prediction

II. HERDING MODEL: FROM CLUSTERS TO PRICE AND VOLUME

A. Agents, actions, and the observation interface

Each period draws $H \sim G(N, p)$, where $p = \rho/(N-1)$ and $N \geq 2$. Nodes are model traders, edges represent coordination, and a connected cluster C contains s_C traders. Each cluster shares $\xi_C \in \{-1, 0, 1\}$ with probabilities $q/2, 1-q, q/2$. Actions are independent across clusters and independent of the graph. Throughout, q is the total activity probability. The earlier code's per-direction probability is $a = q/2$, distinct from bond retention.

Each active trader submits a unit order $u > 0$. A simplified external counterparty absorbs all submitted orders, and the log-price response is linear. The observations are

$$r = \frac{\lambda}{N}\sum_C s_C \xi_C, \qquad V = u\sum_C s_C|\xi_C|, \tag{2}$$
$$S_j = \sum_C s_C^j, \qquad \lambda > 0, \quad 0 < q < 1.$$

Signed price effects can offset, whereas trading activity adds. The volume V is not market-wide buys minus sells, and λ is not calibrated order-book depth. Figure 1 illustrates this observation interface.

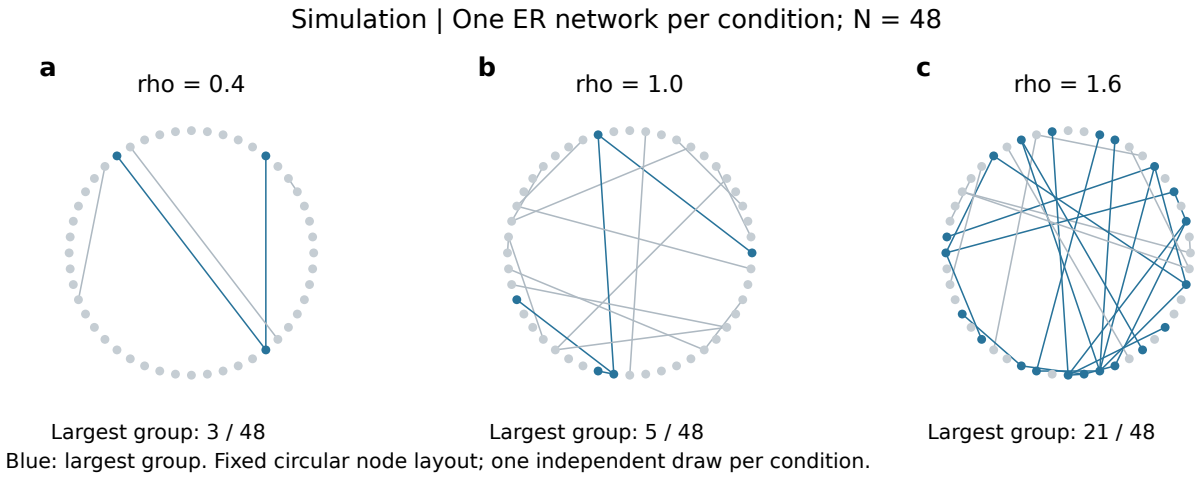


FIG. 1. **MODEL; original F01.** Shared cluster actions generate net price effects and trading activity. The execution rule derives return and volume from the same actions. This represents neither an observed trader network nor a real order book. Definitions appear in docs/AB/model_spec.md; the figure index links original figures to their source tables.

B. Exact moments and their mechanistic use

Zero-mean cluster actions eliminate cross terms in return variance. Conditional mean volume does not depend on the graph. Writing $A = \mathbb{E}S_2$ gives

$$\begin{aligned} \mathbb{E}V &= quN, \\ \mathrm{Var}(r) &= q(\lambda/N)^2 A, \\ \mathrm{Var}(V) &= q(1-q)u^2 A. \end{aligned} \tag{3}$$

Equal mean volume therefore need not imply equal risk. At fixed N, q, u, λ, a larger sum of squared cluster sizes increases return variance. For 100 independent traders, $S_2 = 100$; ten clusters of ten traders give $S_2 = 1000$. Variance differs tenfold in this model example while mean volume remains unchanged. This is a conditional scenario calculation, not a measured market multiplier.

Higher-order relations retain the connection between collective activity and large price changes:

$$\begin{aligned} \mathrm{Cov}(r^2, V) &= u(\lambda/N)^2 q(1-q)\mathbb{E}S_3, \\ \kappa_3(V) &= u^3 q(1-q)(1-2q)\mathbb{E}S_3. \end{aligned} \tag{4}$$

Signed r and V have zero population covariance, while $|r|$ and V are positively associated under this interface. Price direction and magnitude thus require separate analysis. Such an association does not uniquely establish herding in market data.

Conditional on the graph, the joint characteristic function is

$$\mathbb{E}[e^{itr+izV} \mid H] = \prod_C [1 - q + qe^{izus_C}\cos(t\lambda s_C/N)]. \tag{5}$$

It depends only on the multiset of cluster sizes, directly supporting the identification boundary below. For fixed N, $|r| \le \lambda$, so the mathematical tail is bounded. Broad finite-range tails, power-law fits, and size limits remain distinct objects.

Source tables and full methods: `docs/theory/FOUNDATIONS.md`.

III. PERCOLATION, CRITICALITY, AND SCALING

In the infinite sparse ER limit, degrees are Poisson(ρ) and $G_0(z) = G_1(z) = e^{\rho(z-1)}$. The edge-exploration generating function satisfies $H_1(z) = zG_1(H_1(z))$. A giant component emerges when the mean branching number crosses one. Its fraction satisfies

$$P_\infty = 1 - e^{-\rho P_\infty}, \qquad \rho_c = 1. \tag{6}$$

Below threshold, the expected cluster size of a uniformly selected node is $1/(1-\rho)$. In finite systems, S_2/N differs from the statistic excluding one largest component,

$$\chi_f = \frac{\sum_{C \neq C_{\max}} s_C^2}{\sum_{C \neq C_{\max}} s_C} \tag{7}$$

which is recorded as missing when no finite component remains.

We scanned $N = 128, 256, 512$, nine ρ values, and 40 graphs per cell, totaling 1,080 graphs. Figure 2 shows a smooth finite-size connectivity transition. Grid resolution and fluctuations affect peak positions. A sampled finite-size peak is neither an exact threshold estimate nor a market crisis threshold.

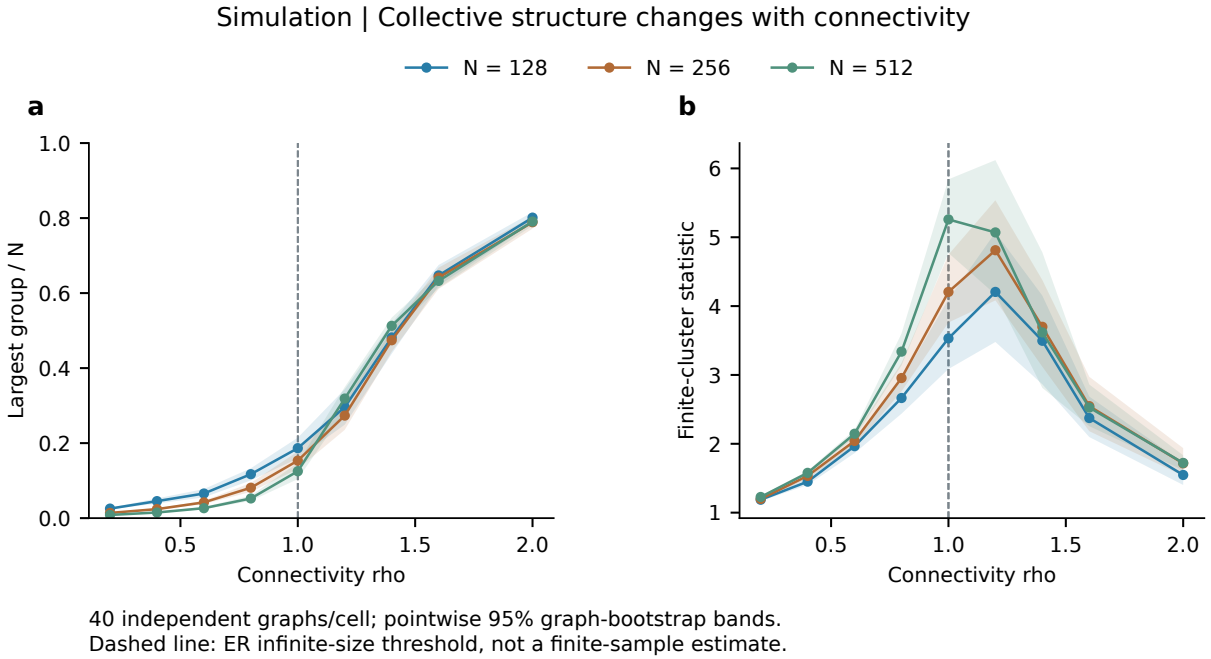


FIG. 2. **SIMULATION; original F03.** Finite-size percolation across three system sizes, with 40 independent graphs per N/ρ cell. Largest-cluster and finite-cluster statistics change with connectivity. The vertical reference is the infinite sparse ER threshold, not an estimated market danger threshold. Source: simulation_study_v1/phase_summary.csv.

The mean-field critical law $n_s \propto s^{-5/2}$ concerns cluster number density. Selecting a node uniformly instead weights cluster sizes by s. Returns sum signed cluster actions; activity, cancellation, and bounded support affect their tails. For density $f(x) \propto x^{-\alpha}$, the CCDF exponent is $\alpha - 1$. These objects cannot share an exponent by assumption.

Five conditions each generated 8,192 periods, totaling 40,960 simulated intervals. Over the prespecified absolute net-order range $m = 4, \ldots, 128$, all three ρ conditions rejected the simple finite-support pure power law. The raw $p = 0.005$ is the resolution floor of 199 parametric-bootstrap draws; Holm adjustment across three tests gives $p = 0.015$. A better relative AIC under some conditions does not imply adequate absolute fit. The implementation follows maximum-likelihood, KS, and alternative-distribution principles[4], adapted to finite integer support.

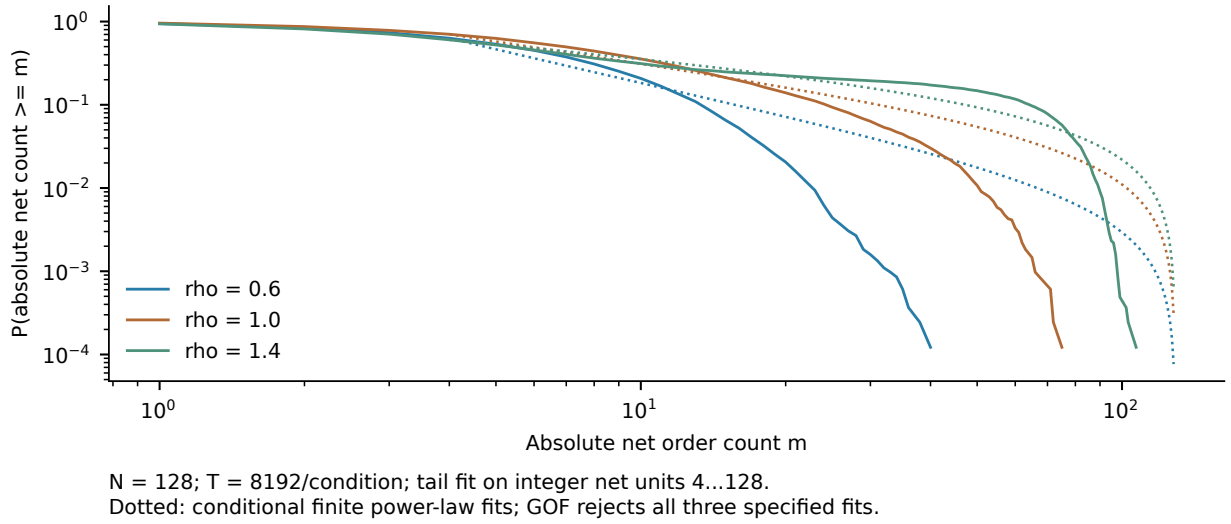


FIG. 3. **SIMULATION; original F05.** Return CCDFs and the specified finite-support tail fits. Log–log appearance and relative AIC do not replace goodness-of-fit testing. The current result rejects this simple pure power law over the stated range, without excluding all thresholds, activity limits, or other models. Appendix A retains cluster and scaling comparisons.

This completes a study of power-law-generating models and scaling that allows an empirical fit to fail. The appendix retains cluster distributions and separate population-size and temporal-aggregation comparisons. No universal return exponent or self-organized criticality is established.

IV. INVERSE INFERENCE: RECOVERABLE PARAMETERS AND LOST STRUCTURE

A. Finite-graph moment inversion

For known N, u under the fixed observation assumptions, population moments give

$$q = \frac{\mathbb{E}V}{uN}, \qquad A_N(\rho) = \frac{\mathrm{Var}(V)}{q(1-q)u^2}, \tag{8}$$
$$\lambda^2 = N^2u^2(1-q)\frac{\mathrm{Var}(r)}{\mathrm{Var}(V)}.$$

The ER mapping $A_N(\rho) = \mathbb{E}S_2$ is strictly increasing and therefore supports inversion. Finite-graph calculations use the classical connectivity recursion[5]:

$$C_s = 1 - \sum_{k=1}^{s-1} \binom{s-1}{k-1} C_k (1-p)^{k(s-k)},$$
$$\pi_{N,s} = \binom{N-1}{s-1} C_s (1-p)^{s(N-s)}, \tag{9}$$
$$A_N(\rho) = N \sum_s s\pi_{N,s}, \qquad C_1 = 1.$$

The recursion is established theory. Our implementation connects it to price–volume inversion, error comparisons, and explicit assumptions; novelty has not been independently established. Calculations used 380 decimal digits, with higher-precision and small-graph enumeration checks.

Ten independently regenerated studies produced 4,920 windows of 256 periods, totaling 1,259,520 simulated intervals. Each study used 252 training, 84 validation, and 84 test windows, plus 72 unseen intermediate-parameter windows. Moment inversion and classical RBF methods share evaluation units. Supervised features exclude seeds and label metadata. Table III and Fig. 4 report the matched-scale results.

TABLE III. Connectivity recovery in the specified ER family: mean MAE over ten independent studies, in units of ρ.

Method	Original grid	Unseen intermediate ρ
Return-only RBF	0.048660	0.045077
Joint RBF	0.027013	0.026882
Volume-only RBF	0.022196	0.036849
Scale-invariant joint RBF	0.032106	0.031013
Moment inversion	0.028372	0.027114

Each study regenerates training, validation, and evaluation windows. Numerical proximity is not formal equivalence. Source: depth_study_v1/summary_metrics.csv.

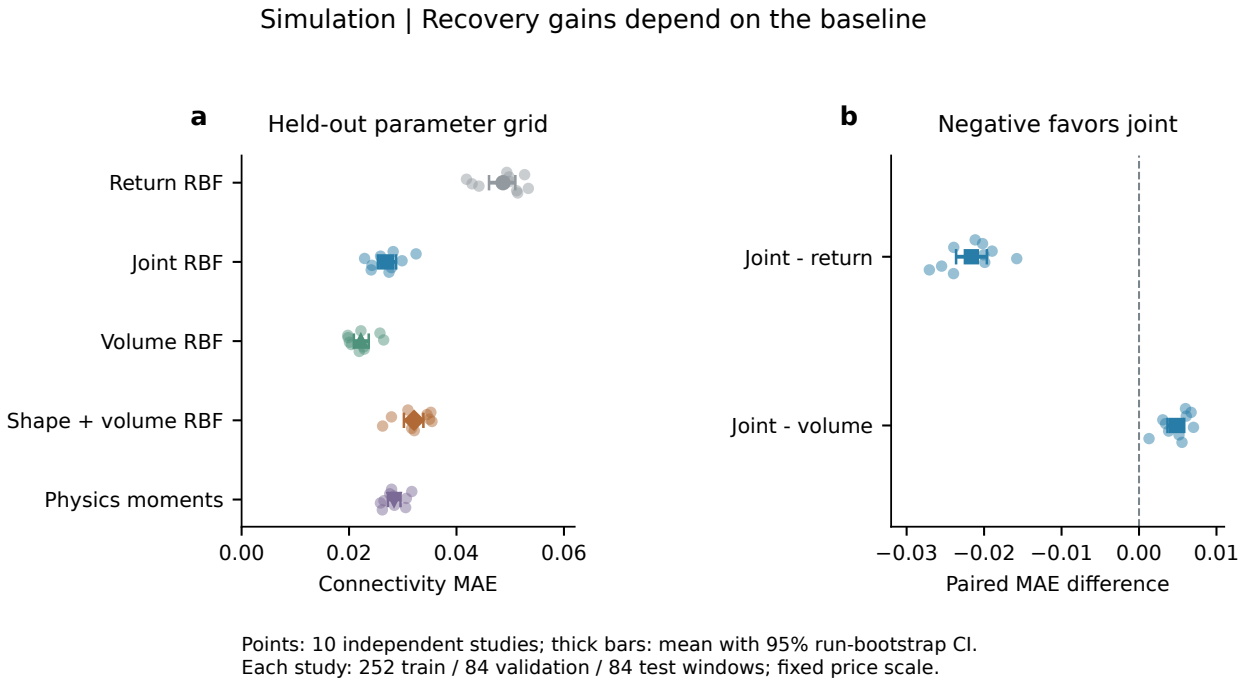


FIG. 4. **SIMULATION; original D01.** Ten parameter-recovery studies regenerate data and retrain models independently. Points represent studies; summaries and intervals describe across-study variation. Joint inputs improve over returns, while volume alone is stronger on the original grid. Paired errors are not market-forecast gains. Table III separately reports unseen intermediate ρ.

On the original grid, joint RBF reduced average MAE by 44.49% relative to return-only RBF. Volume-only RBF performed better than joint RBF in all ten studies. For unseen parameters, moment inversion achieved MAE 0.027114, compared with 0.026882 for joint RBF. Mechanistic formulas can therefore provide interpretable estimators within the specified family. Numerical proximity does not establish statistical equivalence. Appendix B examines scale-invariant inputs and price-response stress without equating estimator failure with non-identifiability.

B. Even complete observations need not reveal internal edges

The chain $\{12, 23, 34\}$ and star $\{12, 13, 14\}$ differ as four-node graphs, yet both form a single cluster of size four. Equation (5) gives identical complete joint price–volume laws. More passive observations of this kind cannot recover the internal edges. This is stronger than failure of a particular classifier. Exact certificates for general cluster mixtures are also included with proofs and rational checks.

Parameter recovery, structural equivalence, and information lost through feature compression require separate treatment. Financial ABM calibration studies likewise distinguish reproducing stylized facts from identifying mechanism parameters[6, 7]. We retain exact counterexamples, known-truth estimation errors, and joint-mechanism tests as separate evidence.

Source tables and full methods: `docs/theory/DEPTH_ADDENDUM.md`.

V. MARKET OBSERVATIONS: COMPATIBILITY AND TEMPORAL STRUCTURE

A. Data and price–volume comparison

The primary market case uses Binance Vision BTCUSDT spot five-minute bars[10]. The 2025 sample contains 105,120 bars and 365 complete UTC days, without filling or deletion. H1 provides calibration; H2 contains the original 184-day compatibility-test period. Returns are adjacent-close log changes, and bar volume is measured in BTC. Five-minute bars still compress the within-bar trade sequence. Figure 5 compares binned joint statistics and response proxies without treating volume as net capital flow.

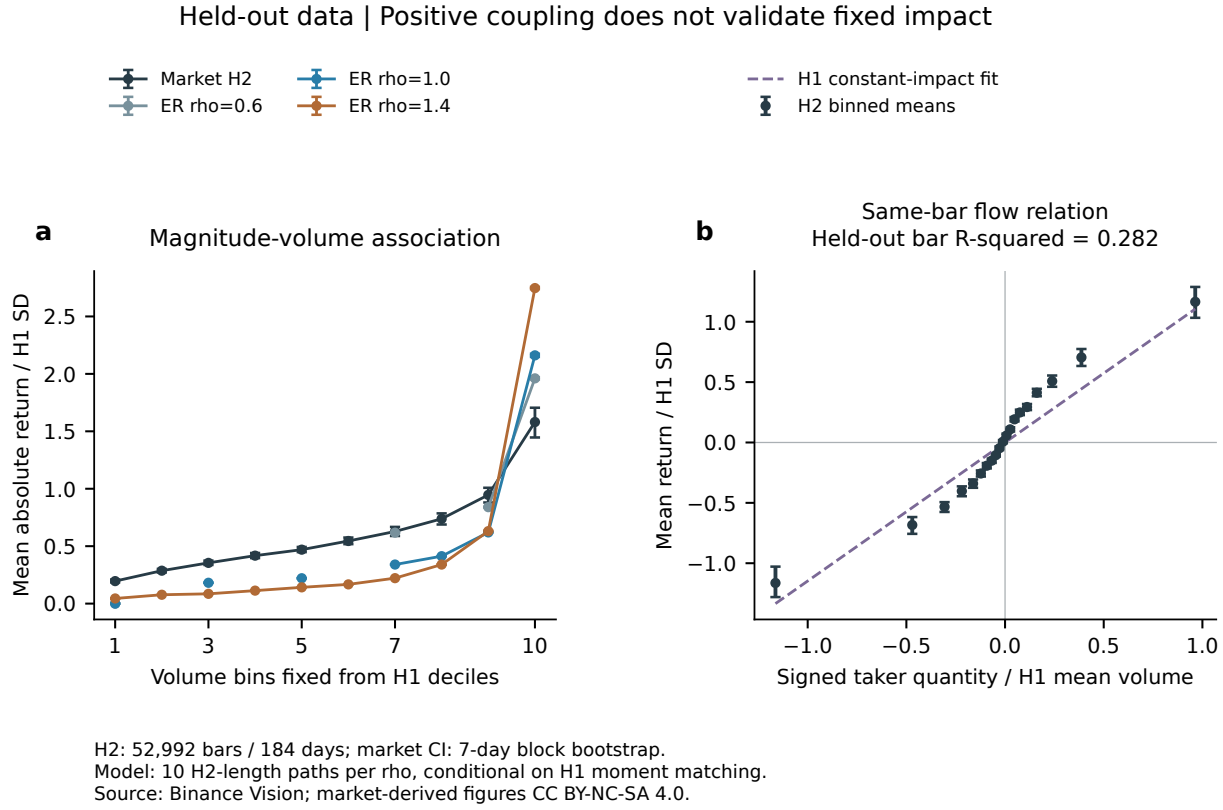


FIG. 5. **REAL DATA + SIMULATION; original R01.** H1-fixed bins compare H2 market observations and model paths. All eligible bars enter the statistics, with empty model bins left missing. H2 magnitude–volume Spearman correlation is approximately 0.498, or 0.481 after seasonal adjustment. Association motivates joint analysis without uniquely identifying herding. Attribution: Binance Vision, CC BY-NC-SA 4.0.

Appendix Fig. 14 compares market and model tails on a calibration-defined scale. Market data contain no true trader counts, clusters, or ρ labels. Simulation-trained outputs therefore cannot directly establish real-market criticality.

B. A joint necessary restriction derived from the mechanism

Eliminating scales from Eq. (4) gives a restriction for any graph distribution, beyond ER:

$$\Delta = \frac{1}{\mathbb{E}V}\left[\frac{\mathrm{Cov}(r^2,V)}{\mathrm{Var}(r)} - \frac{\kappa_3(V)}{\mathrm{Var}(V)}\right] = \frac{\mathbb{E}S_3}{N\mathbb{E}S_2} \in [1/N, 1]. \tag{10}$$

It requires fixed N, q, u, λ, symmetric independent actions, simplified execution, and linear response. Empirical calculations remove constant return drift and retain higher-moment and nonstationarity sensitivities.

For market H2, $\Delta = -8.181$, with an approximate 95% circular seven-day-block interval $[-11.998, -4.712]$. Figure 6 contrasts this estimate with the required positive range. The result challenges the joint fixed-interface assumptions without rejecting every herding mechanism or uniquely identifying an alternative. It provides an operational model audit: similar return tails do not ensure compatible joint observations.

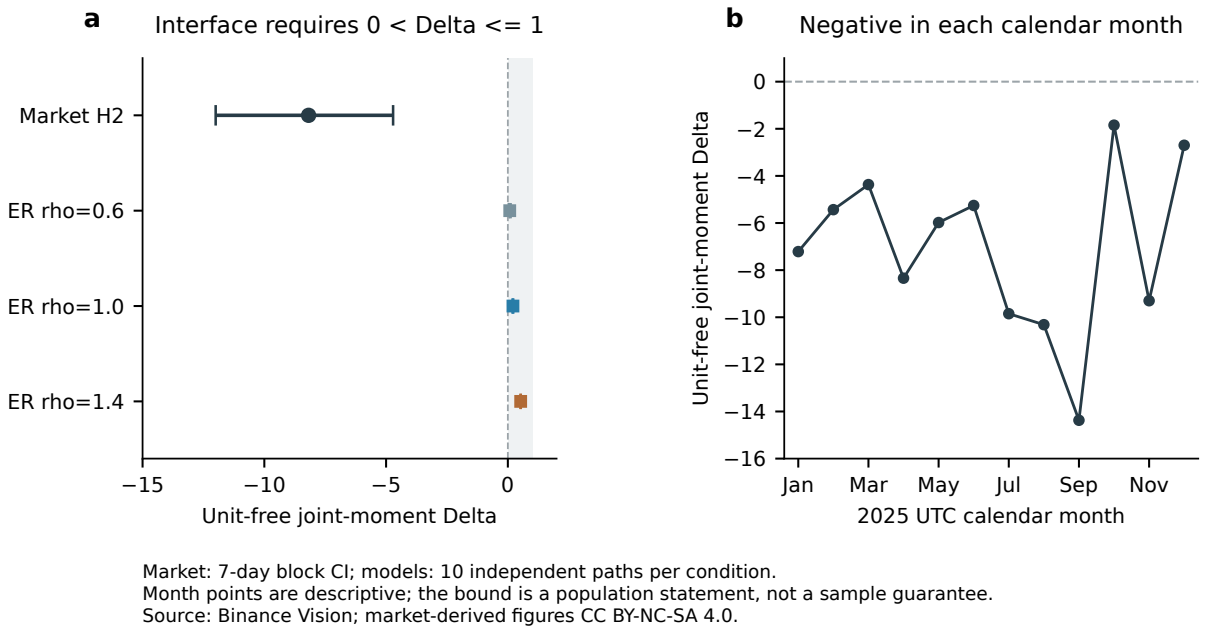


FIG. 6. **REAL DATA + MODEL CHECK; original R04.** The positive model restriction contrasts with negative market estimates. Intervals use approximate circular date-block resampling; model and market points are not independent replications. Monthly and extreme-day sensitivities qualify higher-moment interpretation. The figure does not exclude every herding or network mechanism.

C. Temporal structure supplies additional forecasting information

With fixed parameters and independently redrawn graphs and actions,

$$\mathbb{E}[r_{t+1}^2 \mid \mathcal{F}_t] = q(\lambda/N)^2 \mathbb{E}S_2 \tag{11}$$

is constant. This static baseline does not itself provide a date-varying next-period risk forecast. Market H2 has lag-one magnitude autocorrelation 0.344, signed-return autocorrelation about −0.021, and a one-day aggregation variance ratio about 0.749. Magnitude memory and temporal aggregation are observable features distinct from geometric cluster scaling.

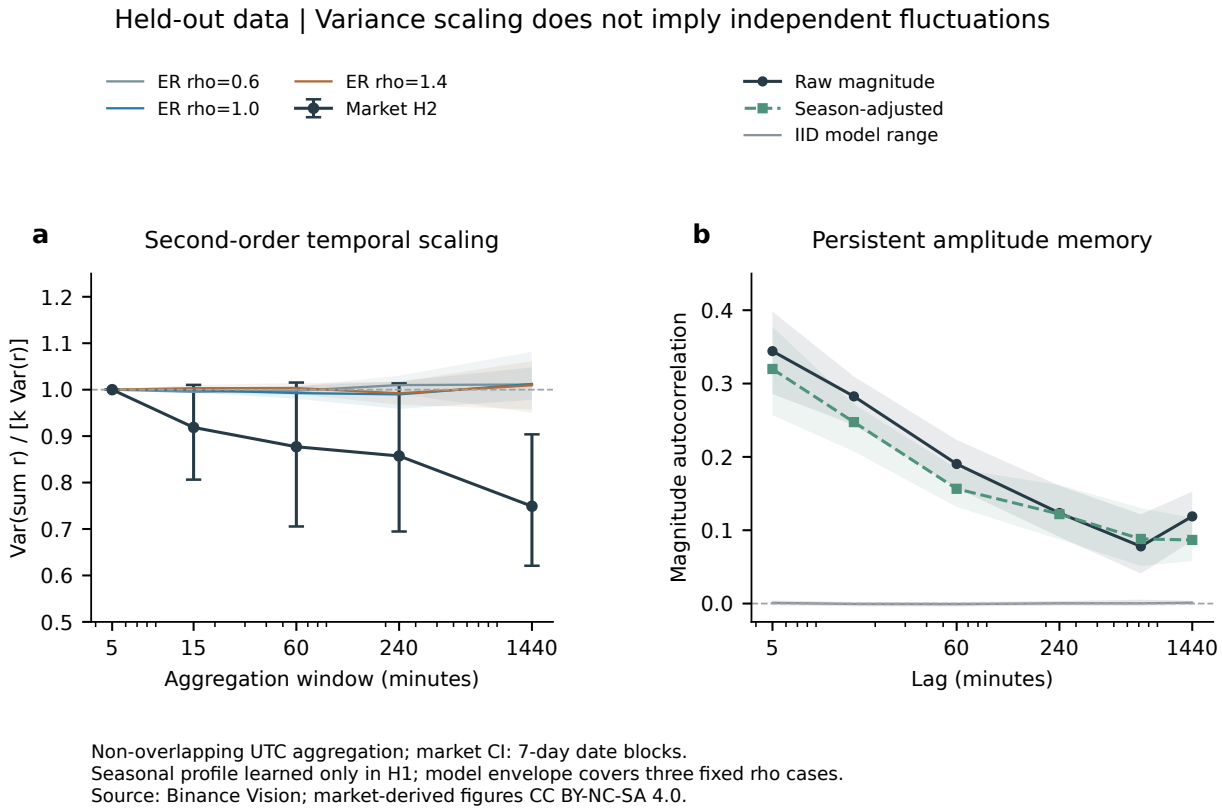


FIG. 7. **REAL DATA + SIMULATION; original R03.** Temporal aggregation and correlation structures differ between market observations and independent simulations. Magnitude memory is present in the market; the static model lacks that mechanism. Variance ratios are neither cluster power-law exponents nor evidence of SOC. Market intervals are date-block approximations; seasonal adjustment and further scales remain in the tables.

VI. NEXT-DAY VOLATILITY FORECASTING AS A PRACTICAL APPLICATION

A. How physics and forecasting are connected

The physics motivates observable activity, joint price–volume statistics, and extreme-concentration measures. Market tests assess the fixed interface and reveal temporal dependence. The forecasting application directly uses these observables and multiscale lags, without true network or cluster labels. Its HAR representation follows the multiscale approach of Corsi[8]; it is not rigorously derived as an ER reduction. Overall forecast skill cannot be assigned to a uniquely identified market herding cause.

After the final five-minute bar of day d closes, the model forecasts the next UTC day:

$$RV_d = \sum_{j=1}^{288} r_{d,j}^2, \quad z_{d+1} = \log RV_{d+1}, \tag{12}$$

$$\widehat{z}_{d+1} = \bar{z}_{\text{fit}} + \widehat{w}^\top \widetilde{x}_d,$$
$$\widehat{w} = \arg\min_w \sum_{d \in \text{fit}} (z_{d+1} - \bar{z}_{\text{fit}} - w^\top \widetilde{x}_d)^2 + \alpha_R \|w\|^2. \tag{13}$$

Standardization uses the fitting period only. The ridge penalty α_R differs from a tail exponent. The displayed $100\sqrt{RV}$ measures intraday volatility intensity, without specifying the direction of the close-to-close return.

TABLE IV. The full model's ten inputs, all available at the forecast origin.

Group	Definition	Count
HAR	$\log RV_d$; log mean RV over the past 7 days; log mean RV over the past 28 days	3
Activity	Log total daily BTC volume; log ratio of daily volume to its past-7-day mean	2
Coupling	Bar-level Spearman$(\lvert r\rvert, v)$; $\overline{r_c^2 v_c}/(\overline{r_c^2}\,\bar{v})$; absolute summed signed-taker volume divided by daily volume	3
Tail	$\log[288 \sum r_j^4/(\sum r_j^2)^2]$; $\max r_j^2/\sum r_j^2$	2

r_c, v_c are demeaned within a day. Signed activity derives from venue taker fields, not full order-book OFI or market-wide buy–sell imbalance. Tail inputs measure finite-sample shape and concentration, not power-law exponents. The 1/7/28-day horizons adapt the setup to a 24/7 market.

The original protocol used 337 eligible training targets in 2025 and 90 validation targets in 2026 Q1. After selection, one refit used all 427 targets. Parameters remained fixed across 183 evaluation dates from April 1 to September 30, 2026. Observed test-day data can become later lagged inputs, but do not enter refitting or renewed selection. Figure 8 retains every test date.

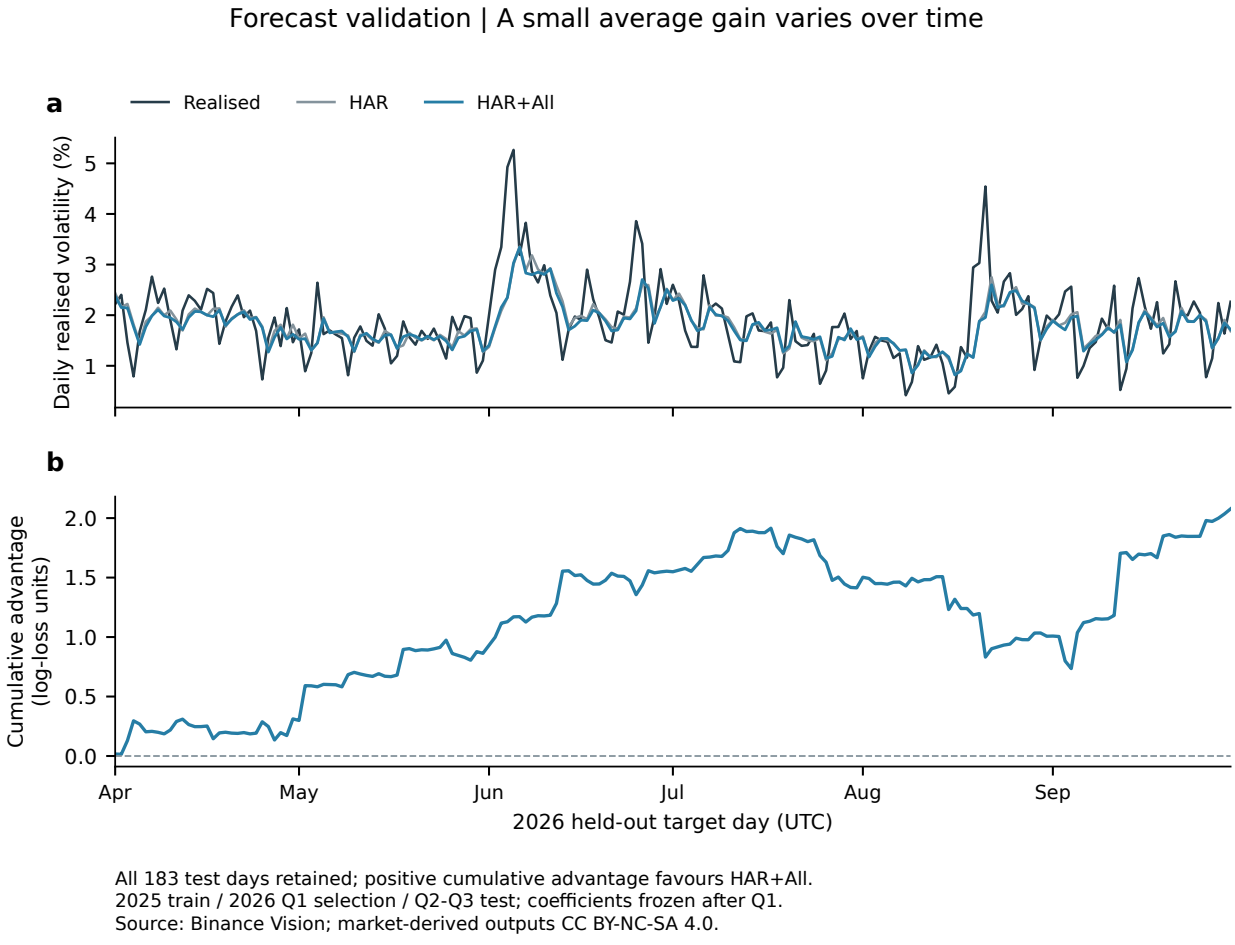


FIG. 8. **REAL DATA; original P01.** Forecasts and realizations across all original 183 dates. Volatility is $100\sqrt{RV}$. The lower panel accumulates loss advantage over HAR, not investment returns. Point forecasts miss some extreme movements; the plot contains no calibrated future prediction intervals. Binance Vision, CC BY-NC-SA 4.0.

B. Overall skill and incremental value are separate

The primary loss and relative skill are

$$L_{\log} = D^{-1} \sum_d (\log RV_d - \widehat{z}_d)^2,$$
$$\text{Skill}_B = 100(1 - L_{\text{model}}/L_B). \tag{14}$$

Table V and Fig. 9 place simple and strong references together. The full model has positive historical skill over a constant, persistence, and EWMA. HAR alone already improves over persistence by 29.61%. The additional 2.263% point gain over HAR has an interval crossing zero. The original primary comparison has Holm-

adjusted $p = 0.3948$, without confirmed incremental advantage. HAR+All adds return shape as well as trading information, so its incremental change is not volume-only.

TABLE V. Original next-day log-RV evaluation: 183 BTCUSDT dates.

Reference/model	log-MSE	All loss reduction (%)	Approx. 95% CI (%)
Fit-period mean	0.793999	38.26	[24.65, 48.97]
Persistence	0.712561	31.20	[21.76, 38.62]
EWMA	0.606913	19.23	[13.22, 24.99]
HAR reference	0.501565	2.26	[-0.11, 4.56]
HAR+All	0.490213	—	—

All denotes the original HAR+All. Intervals use 4,999 paired circular seven-day blocks, unadjusted, in a post-analysis overall-skill reassessment. The original unconfirmed incremental result is retained. Source: predictive_utility_v1.

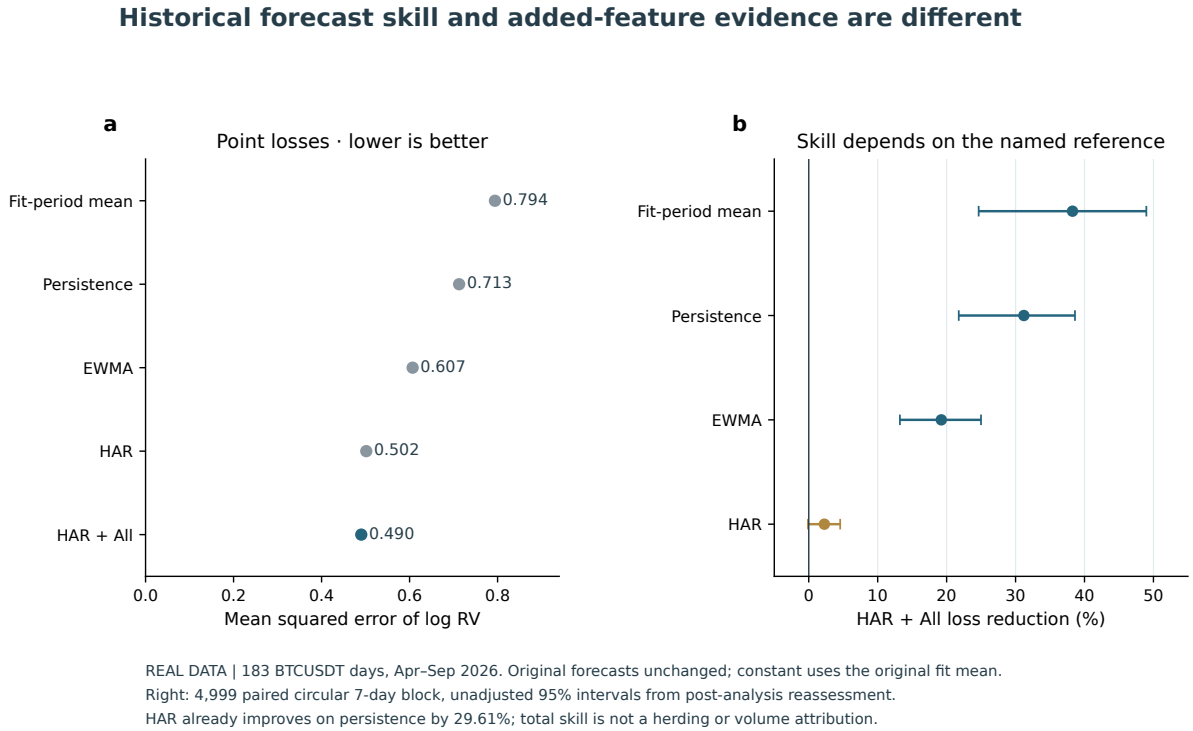


FIG. 9. **REAL DATA; synthesis figure J02.** Left: log-MSE levels. Right: full-model skill against four references with approximate intervals. The constant reference and overall-skill reassessment followed inspection of the original results; they are not new preregistered validation. All nine original models, three losses, and 1/7/14-day-block sensitivities remain available. Crossing zero does not establish equivalence to HAR.

This is a reproducible volatility-analysis application with next-day estimates and a complete historical error record. It can supply inputs to exposure-risk characterization without requiring a uniquely superior algorithm. Risk-capital adequacy, automated-allocation returns, and real-time deployment benefits remain unvalidated. Subsequent asset and time tests qualify generalization of added price–volume information, as detailed in Appendix C.

VII. OMITTED RISK, STRATEGY REDUNDANCY, AND THE INTERACTIVE TOOL

A. How much sample risk can common movement conceal?

The US-equity study contains nine sector ETFs, SPY and QQQ references, and 46,530 daily rows through October 23, 2026. Financial association networks have established precedents[9]; our sector graph represents observed co-movement, not trader interactions. The original design uses 2010–2017 for training, 2018–2019 for selection, 2020–2025 for historical evaluation, and 2026

for confirmation.

For within-window return covariance $\widehat{\Sigma}$ and weights w,

$$\widehat{v} = w^{\top}\widehat{\Sigma}w = \underbrace{\sum_i w_i^2\widehat{\sigma}_i^2}_{v_{\text{self}}} + \underbrace{2\sum_{i<j} w_i w_j \widehat{\sigma}_{ij}}_{v_{\text{collective}}} . \qquad (15)$$

Annualized volatility is $100\sqrt{252\widehat{v}}$. For nine equally weighted sectors in the 63-day window ending March 31, 2020, the full estimate is 57.73%. Retaining only diagonal terms gives 20.62%, a ratio of 2.80. Collective terms account for 87.24% of full estimated variance. These are same-window decompositions, not future forecasting gains. Figure 10 retains the longer rolling history.

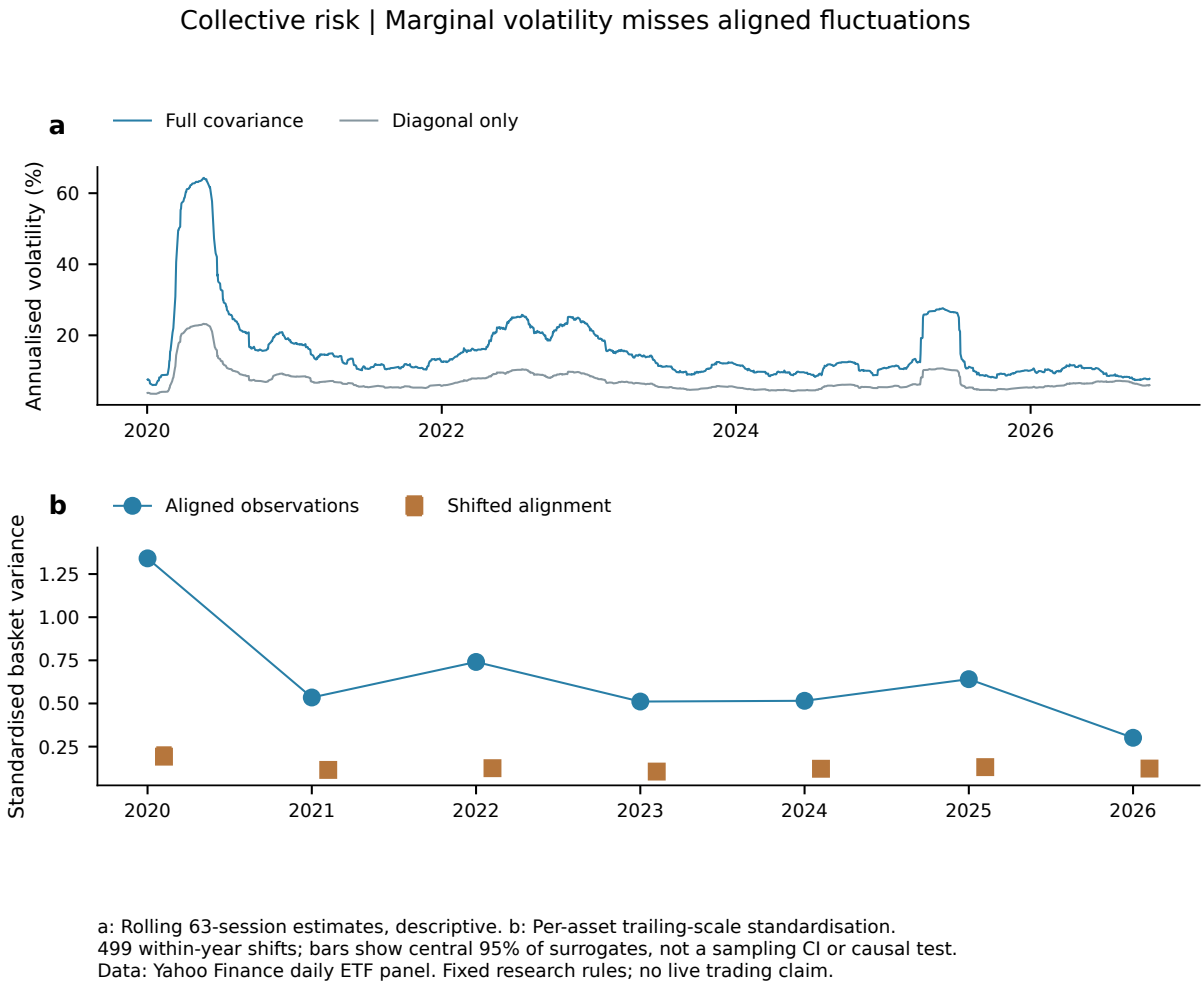


FIG. 10. **REAL DATA; original U02.** Sector-risk decomposition and conditional shifts of cross-asset date alignment. Numeric risk uses the full covariance matrix, independently of the displayed edge threshold. Shift ranges are neither sampling confidence intervals nor causal p values. The 57.73%/20.62% example comes from risk_lab_v1/risk_snapshots.csv.

The existing tool lets users change dates, 63/126-day windows, and holdings. It also evaluates $\Sigma_\eta = (1-\eta)\,\mathrm{diag}(\widehat{\Sigma}) + \eta\widehat{\Sigma}$. This positive-semidefinite interpolation is a conditional counterfactual, not a newly observed market. The tool makes conventional covariance analysis operational, without claiming a physics-specific algorithm.

B. Strategy variants are not independent risk sources

The six fixed proxies use momentum at 63/126 days, reversal at 5/21 days, and low volatility at 21/63 days. Dependence is assessed primarily through daily net P&L increments. The low-volatility pair has daily correlations 0.921/0.795 in the historical/2026 periods, versus cumulative-curve correlations around 0.982/0.985. Cumulative curves therefore cannot replace daily-risk analysis. Figure 11 separates shared exposures and cross-asset

terms.

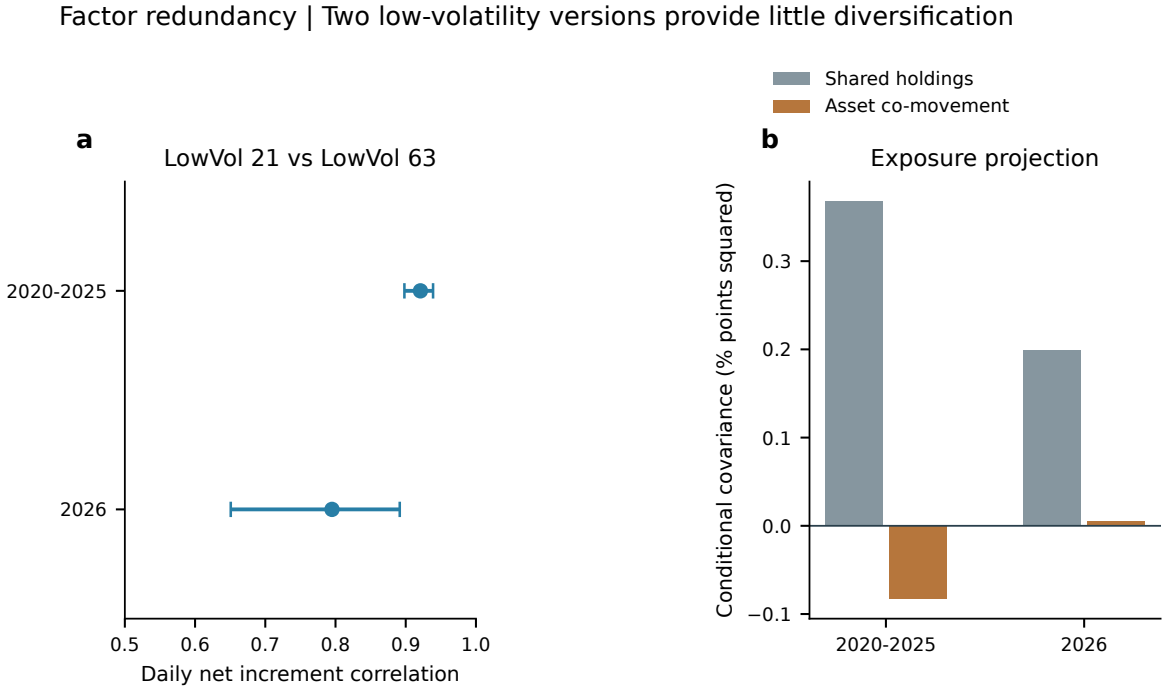


FIG. 11. **REAL DATA; original U05.** Redundancy of fixed low-volatility variants and conditional risk decomposition under known exposures. Date-block intervals for daily correlations differ from cumulative-curve comparisons. For two equal-risk strategies, $2/(1+\rho)$ is a covariance-based equivalent count, not a count of independent alphas. Cross-asset terms can offset long–short risk.

The default screening rule uses only 2011–2017 daily net increments and turnover. It ranks candidates by turnover and flags positive correlations ≥ 0.80. Six candidates become five representatives, reducing their count by 16.7%. Evaluation-period statistical reconstruction of the flagged strategy gives $R^2 = 83.9\%/64.1\%$. The rule was proposed after evaluation data had been inspected and remains exploratory. Reconstruction R^2 is neither retained alpha nor an executable hedge return.

TABLE VI. Default screening from six strategies to five representatives.

Period	Cumulative net P&L (%)	Annual P&L volatility (%)	Daily turnover (%)
2020–2025	-21.68→-20.77	4.35→4.33	7.46→8.67
2026 confirmation	-6.44→-5.57	3.84→3.99	8.03→9.05

Fixed reference capital with additive P&L; 5 bp one-way transaction cost and assumed 100 bp annual borrowing cost. Screening is post-analysis exploration, not a new blind test. Fewer representatives need not reduce turnover or risk. Source: risk_lab_v1/subset_metrics.csv.

Table VI shows higher combined turnover after removing one candidate, together with higher 2026 volatility. The useful workflow flags candidates, then checks combined holdings, costs, and exposures before deciding what to retain. Redundancy is not an automatic deletion instruction. All 63 nonempty subsets can be inspected; the tool does not automatically select the evaluation-period winner.

C. How the mechanism enters the tool

With additional assumptions of common loading v, independent noise covariance D, and fixed strategy exposures W,

$$\Sigma_Z = Kvv^\top + D, \quad K = q(\lambda/N)^2\mathbb{E}S_2,$$
$$\Sigma_{\text{strategy}} = W\Sigma_Z W^\top. \tag{16}$$

The equation connects a coordination-generated common mode to asset and strategy risk. Common news, holdings, and heteroskedasticity can create similar correlations, so the cause is not uniquely identified. The fixed-W relation is also not an exact full-sample identity for dynamic strategy P&L.

The existing single-file tool has four modules: omitted risk, factor screening, allocation comparison, and physical mechanisms. Its mechanism module varies ρ, q, redraws graphs, and simulates actions while displaying cluster moments and model risk. The common-loading scenario is explicitly not fitted to US equities. Tool files, settings, metrics, and prior interface checks are included.

Source tables and full methods: `web/advanced/index.html`.

VIII. INTEGRATED CONTRIBUTION AND SCOPE OF USE

The physics contributes testable generative relations, conditional inversion, and identification boundaries. Market observations distinguish generating a phenomenon from adequately explaining the observed interface. The volatility application exploits temporal information, while the risk tool provides operational diagnostics and comparisons. Together they form a mechanism–observation–test–application workflow, without assigning every contribution to one performance number.

Supported uses within this study include conditional stress scenarios, model audits, historical volatility forecasting, and redundancy/cost/exposure checks. True market ρ, crash dates, universal power laws, stable cross-market superiority, and live investment profits are not established. The appendices retain uncertain or negative state, covariance-matched tail, and exit-feedback results. Those results are not retuned into successes. QML was not run and is not a completed result.

The public research record includes the interactive website, three executable notebooks and the shared code repository.

A. Research outlook

Future work may explore quantum models and quantum machine learning in relation to collective dynamics, price–volume information and risk. Whether, how and when to pursue these directions will depend on their suitability for the research questions, the availability of a testable formulation and the value of comparison with classical methods. No implementation schedule has been set.

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[10] Binance Vision, Public Market Data; terms and checksum snapshots are included in the package. Market-derived outputs: CC BY-NC-SA 4.0, attribution Binance Vision.

Appendix A: Complete tail and scaling objects

Cluster number density, node-weighted cluster size, and return distributions are different observables. Figures 12 and 13 retain cluster evidence and separate population size from temporal aggregation. Independent-model second-order aggregation differs from market memory. Similar-looking curves do not establish a universal exponent.

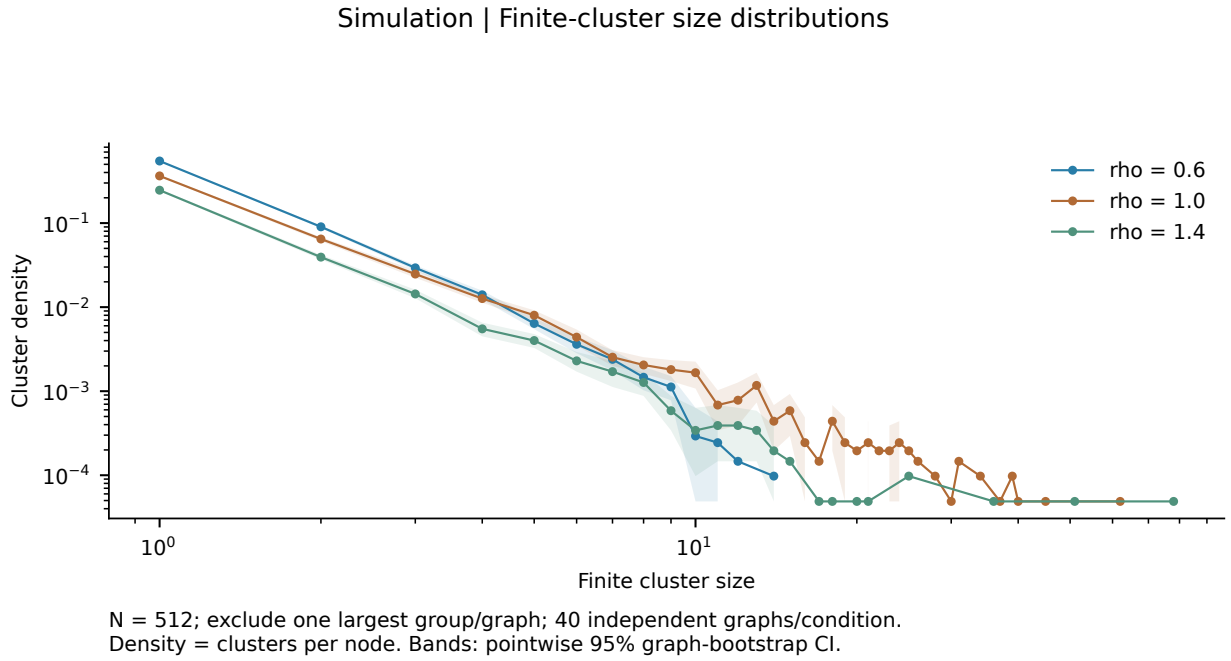


FIG. 12. **SIMULATION; original F04.** Finite-cluster number density at the specified system size, with pointwise graph-bootstrap bands. Number density differs from node-weighted cluster size and cannot directly supply a return-density or CCDF exponent. Raw cluster sizes and all system sizes remain in the source tables.

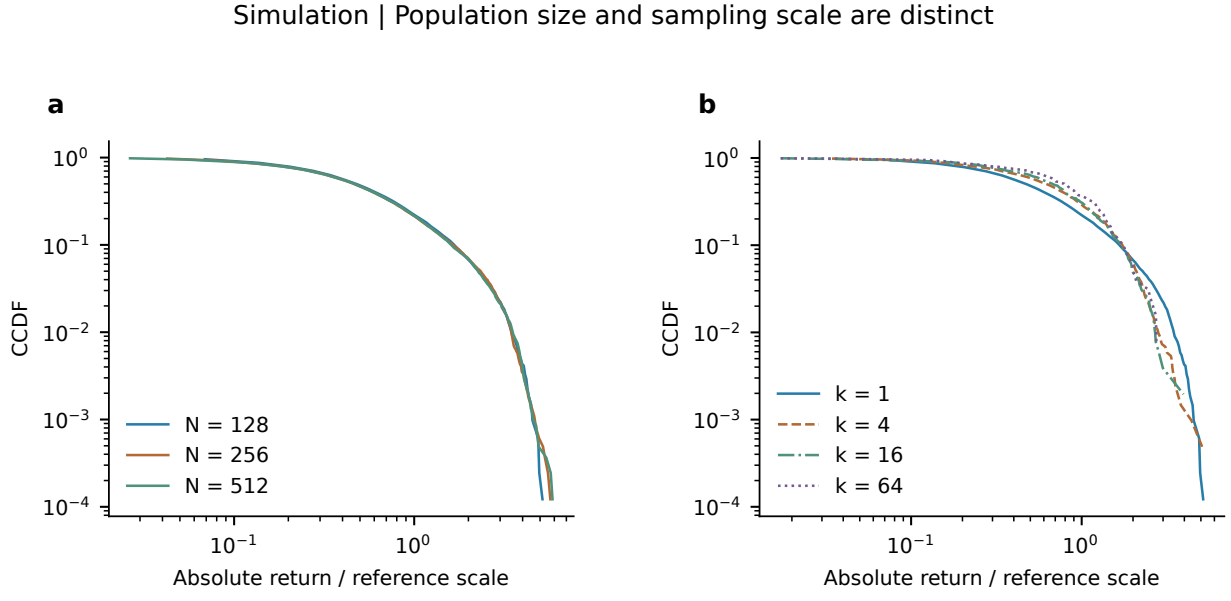


FIG. 13. **SIMULATION; original F06.** Separate population-size and temporal-aggregation comparisons. Standardized shapes aid comparison without establishing universality by appearance. Temporal blocks are nonoverlapping, and all conditions remain available.

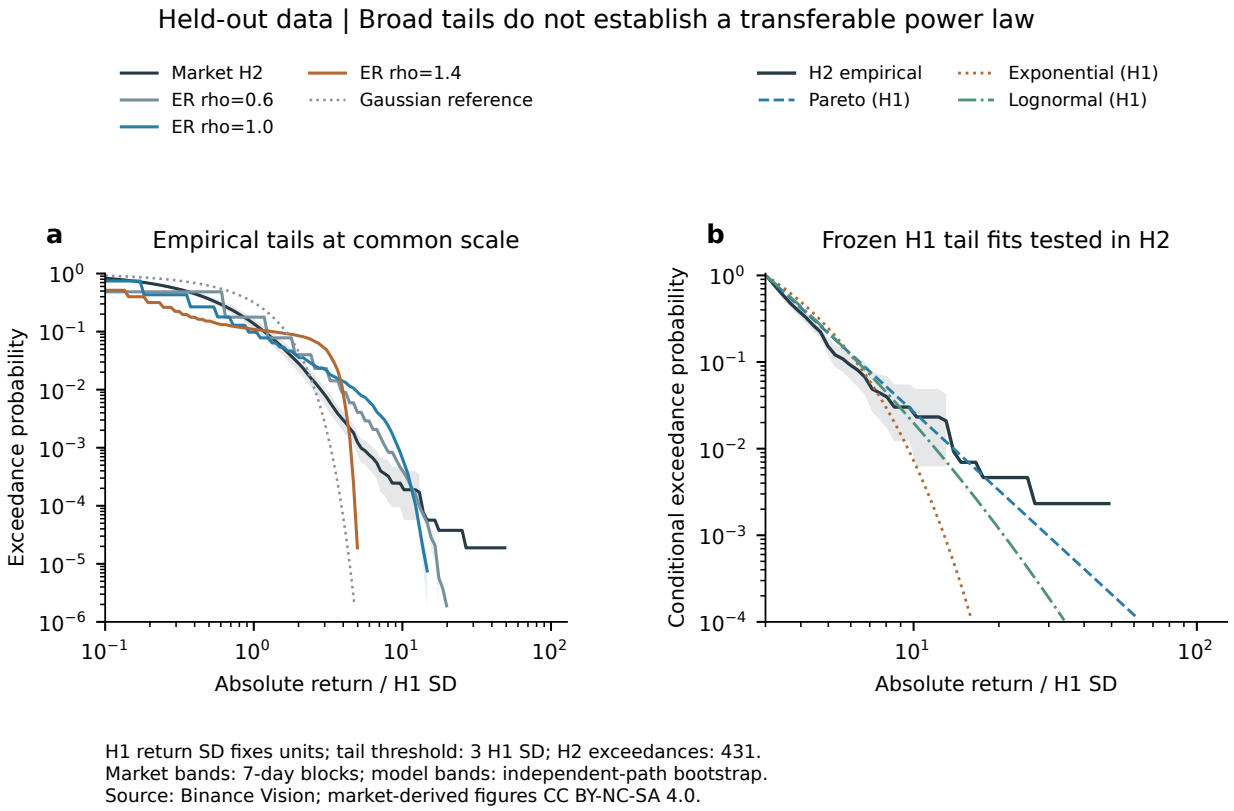


FIG. 14. **REAL DATA + SIMULATION; original R02.** H1-scaled H2 return CCDFs, conditional tails, and continuous alternatives. Zero exceedances and zero interval endpoints are omitted only from log display, not the data. Continuous market-tail estimators are not conflated with discrete simulation estimators. Binance Vision, CC BY-NC-SA 4.0.

Figure 14 complements joint observations, compatibility, and memory tests rather than selecting only similar-looking tails. All 28 market CSVs and original reproduction records are retained. Source maps distinguish model seeds from observable inputs.

Appendix B: Scale robustness, exact proofs, and input controls

Each window undergoes $r \mapsto br$, with $b = 0.5, 1, 2$, while V remains unchanged. These are paired transformations of the same windows. Doubling the price response increases raw joint-RBF MAE from 0.027013 to 0.258133. The scale-invariant representation and moment method remain at 0.032106 and 0.028372, respectively. Figure 15 shows this representation trade-off. It establishes neither robustness to every market change nor non-identifiability from estimator failure.

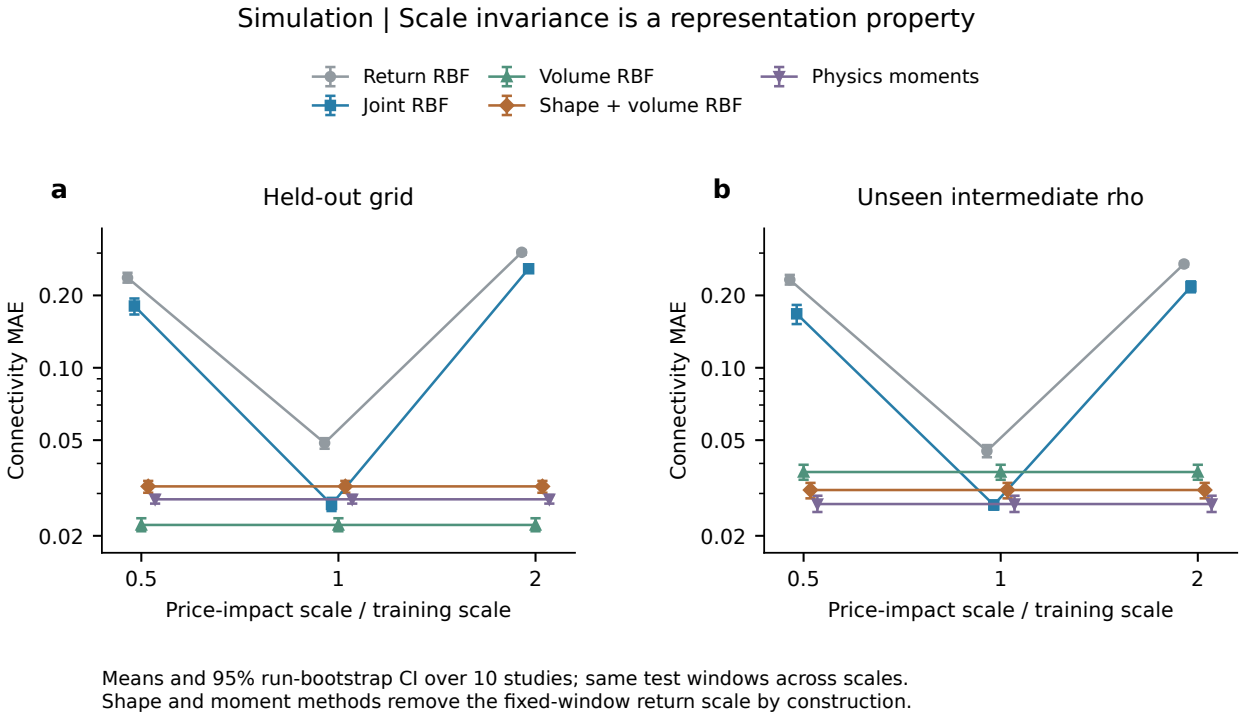


FIG. 15. **SIMULATION; original D02.** Price-response stress across ten studies, reported separately for grid and unseen intermediate parameters. Moment and scale-invariant representations remain unchanged under the stated window-level transformation, without being uniformly best at the original scale. Intervals resample studies; three scales are not three independent datasets.

A minimal persistent-state extension keeps $N = 128, \rho = 1$ and independent graphs, while activity q_t or response λ_t follows an exogenous temporal state. Four arms each use ten paths of 8,192 periods, totaling 327,680 observations. Population mean volume and return second moments match across arms. Persistent activity produces magnitude and volume memory; persistent independent response can produce magnitude memory with nearly absent volume memory. Shuffling complete (r, V) pairs preserves the same-period empirical joint distribution while removing temporal memory. Figure 16 separates these informational roles.

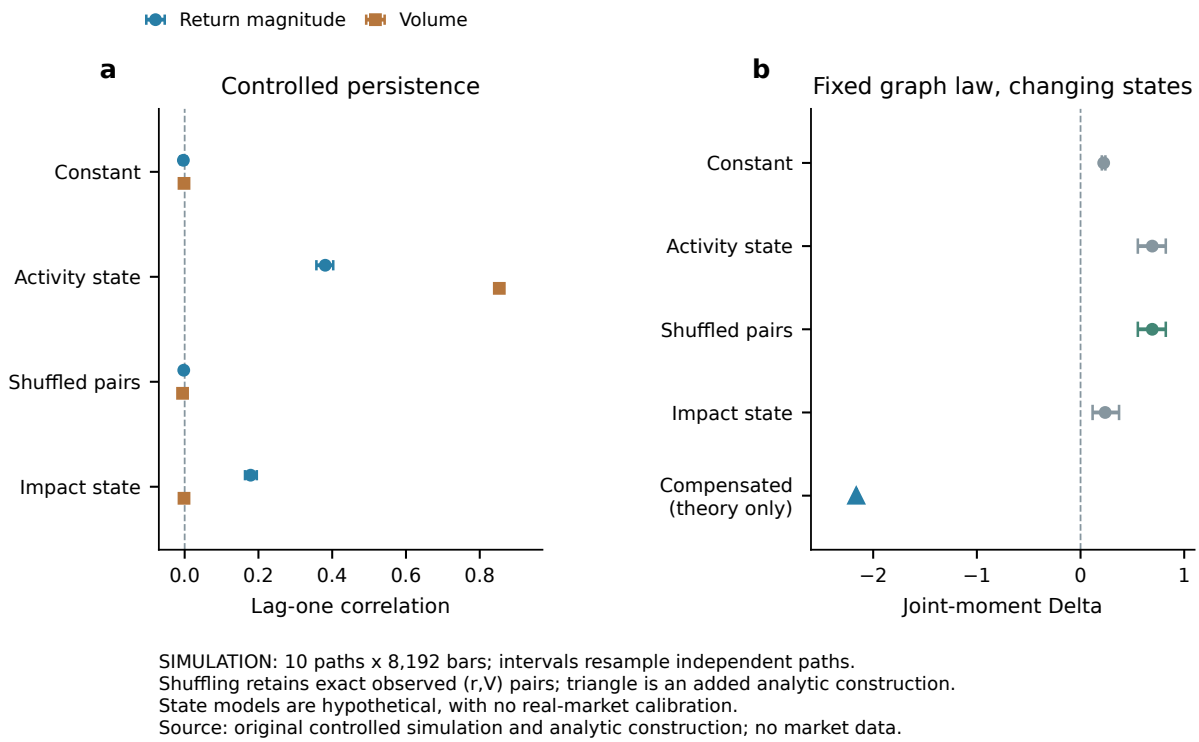


FIG. 16. **SIMULATION + MODEL CHECK; original V01.** Controlled persistence and pair-shuffling comparisons under the same graph law. Path-level intervals differ from market date-block intervals. Compensation is post-analysis population theory, not a fifth prespecified forecasting model. Exogenous persistence is not endogenous adaptation or SOC. Sources: state_mechanism_v2/summary.csv and compensation_theory.json.

Let $m_k = \mathbb{E}(q^k)$ and $A = \mathbb{E}S_2$, with activity independent of the graph. The total-variance identity gives

$$\mathrm{Var}(V) = u^2[(m_1 - m_2)A + N^2(m_2 - m_1^2)]. \qquad \text{(B1)}$$

The second term arises from changing activity, preventing interpretation of all volume variance as cluster structure. Another post-analysis construction sets $\lambda(q)^2 = \lambda_0^2\bar{q}/q$, keeping $\mathbb{E}[r_{t+1}^2 \mid \mathcal{F}_t] = (\lambda_0/N)^2\bar{q}A$ constant. Volume may reveal activity states without improving this forecasting target. Its population $\Delta \approx -2.16348$ shows that the positive fixed-interface restriction does not extend to all herding mechanisms. It neither fits the market value -8.18 nor establishes real compensation. Full assumptions and exact checks appear in STATE_MECHANISM_v2.md.

Theory files contain proofs of Eqs. (3)–(4), monotone identification, and chain/star and mixture equivalence. Rational enumeration and exact certificates check implementations. The original 480 theory checks and ten-study refit records are included. Check counts describe implementation evidence, not novelty or proof strength. D03 retains the marginal-preserving permutation test: marginal fit cannot replace joint-dependence testing, and that test is not a criticality detector.

Appendix C: Supplementary forecasts and unconfirmed extensions

TABLE VII. Supplementary forecast losses, without replacing the primary metric.

Model	QLIKE	Volatility MAE (percentage points)
HAR	0.234214	0.440987
HAR+All	0.233933	0.433735

Volatility is $100\sqrt{RV}$. Exponentiating a log prediction need not yield a conditional arithmetic mean.

Table VII retains supplementary losses. Monthly point gains over HAR are 2.90%, 5.41%, 4.99%, −1.12%, −1.71%, and 5.15%. The negative months remain visible. A post-analysis comparison against HAR+Tail gives a 1.535% point gain, again with an interval crossing zero. Only eight high-volatility events occur in the final sam-

ple, insufficient for a crash-warning claim.

The subsequent state study adds an external ETH evaluation and 25 new dates for each of BTC and ETH. BTC's new-time primary QLIKE gain is 2.476%, with approximate 95% interval $[-0.407, 4.815]$%. Validation selects Off for the ETH trading channel. Complex Gamma specifications do not consistently beat simple HAR, as retained in Fig. 17. There are 25 distinct dates, not 50 independent observations across two assets.

On the same short BTC period, supplementary log-MSE improves by 3.170% against matched return-only Gamma. Its approximate 95% interval is $[1.236, 4.733]$%, with approximate Holm $p = 0.0276$ across that three-test supplementary family. This positive result does not replace primary QLIKE or external-asset checks, or establish superiority to HAR. Twenty-five dates provide only about 3.6 seven-day blocks, requiring a strong dependence approximation.

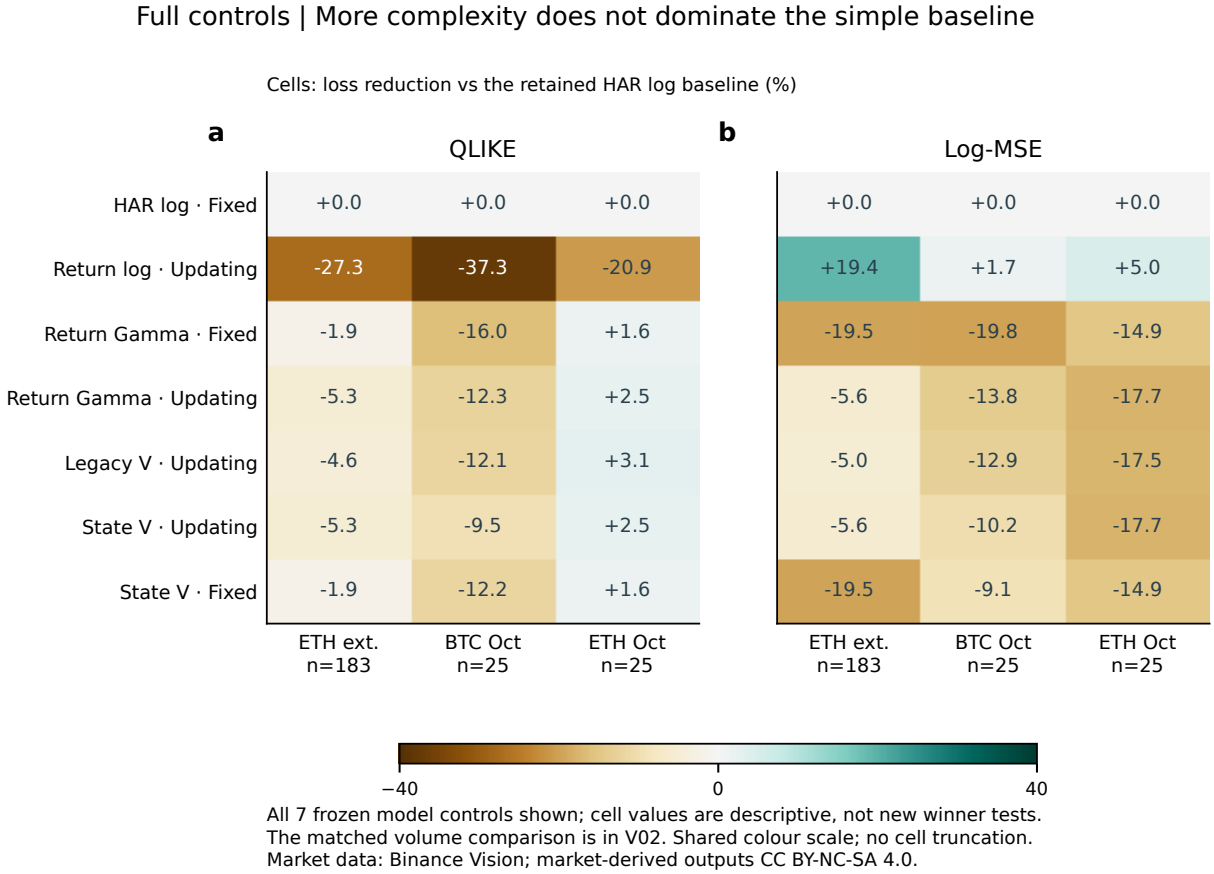


FIG. 17. **REAL DATA; original V03.** Complete comparison with retained simple HAR and other controls. Previously inspected BTC periods are development information; new assets and new time remain separate. Off is a validation-selected return-only fallback, not proof that volume has no information under every model. Binance Vision, CC BY-NC-SA 4.0.

Covariance-matched tail tests hold information and covariance fixed, comparing nine methods on 38,070 rows from nine external stocks. Historical tail improvement is 0.557%, with Holm $p = 0.0772$; the 2026 point change is −1.797%. There are 32 historical co-crash events and zero in 2026, restricting the latter evaluation to false alarms. Figure 18 and the complete controls do not establish stable market increment. External-asset holdout is not a new future-time blind test.

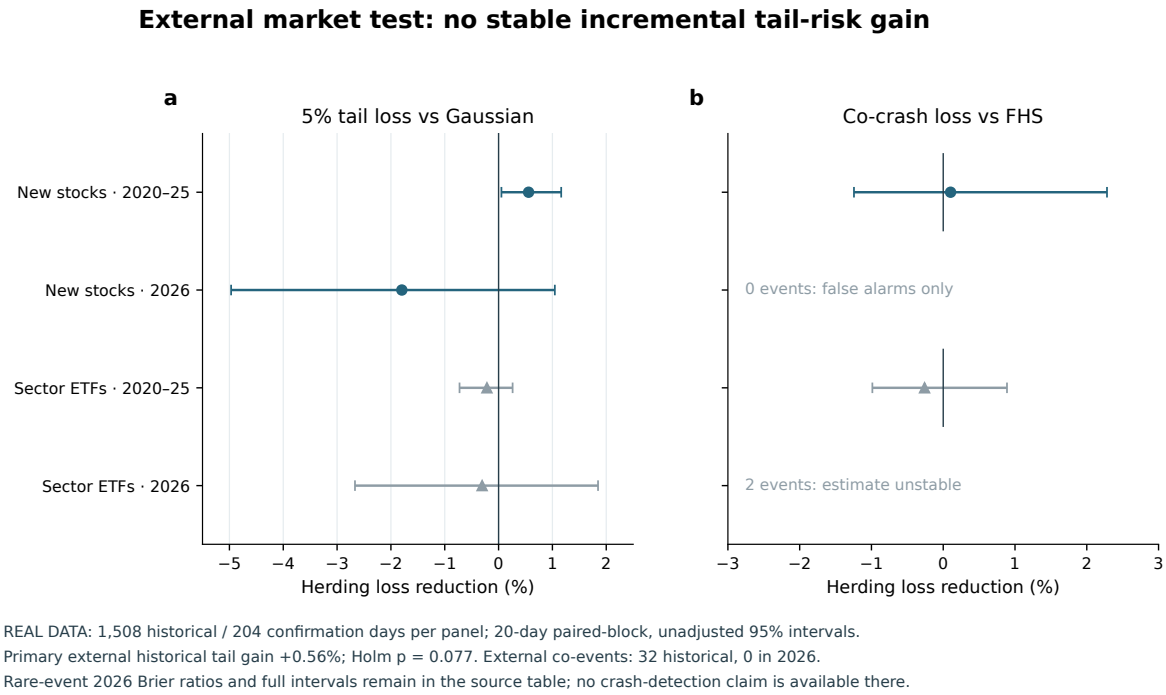


FIG. 18. **REAL DATA; original T03.** Tail and joint-downside increment under matched information and covariance. Strong statistical references, shrinkage, integration checks, and criticality ablations remain included. Distinguishable simulated mechanisms need not yield stable market gains. Zero 2026 events limit joint-downside interpretation.

Appendix D: Stopped capacity extension and volume trade-offs

Six strategies, nine ETFs, 190 monthly snapshots, four allocations, and three depth levels produce 2,280 conditional states. Volume-only, one-round impact, and multiround-feedback capacity bounds coincide in every state, giving 0% extra reduction in Fig. 19. The maximum initial loss, 1.79158%, is below the fixed 2% trigger. A certificate also verifies nonactivation at smaller scales. The branch stopped at its original gate, without changing thresholds or assets to obtain gains.

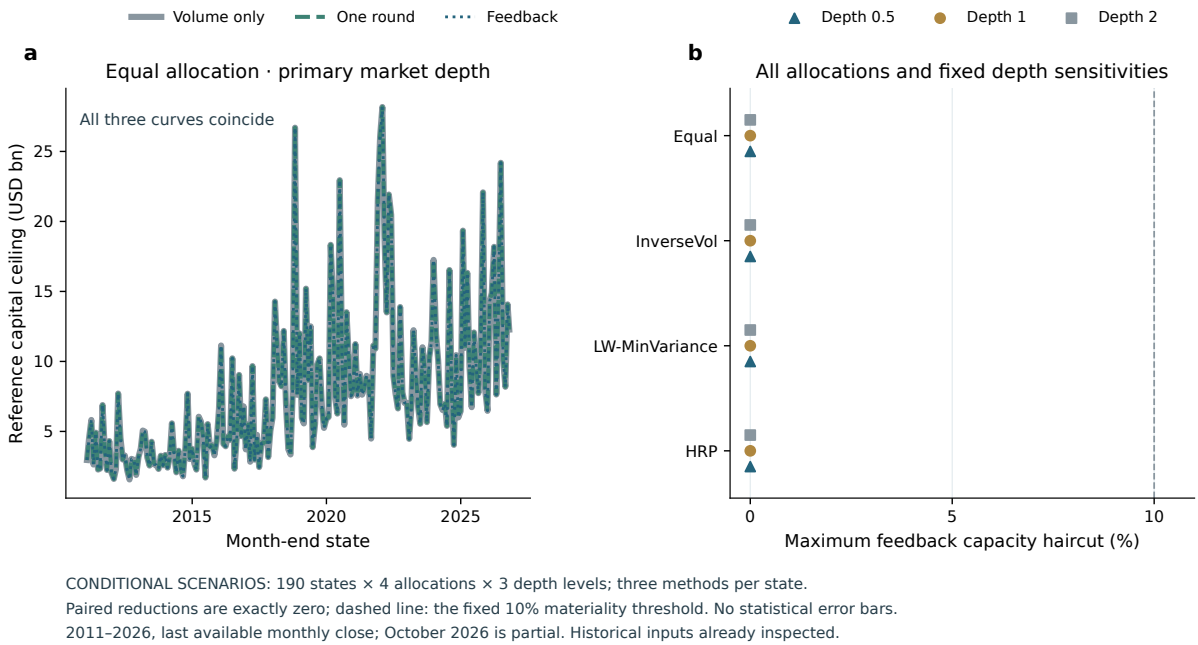


FIG. 19. **CONDITIONAL MODEL; original C01.** Coincident bounds under a fixed account self-impact scenario. Historical market data are conditional inputs; execution data do not calibrate the feedback rule. The model includes no exogenous shock or outside-fund trading and does not measure executable market capacity. All 2,280 states are retained; this is not a significance test.

A conventional volume proxy still exposes a useful trade-off. Among 189 Equal-allocation historical replays with three complete future days, mean-volume sizing exceeds the participation proxy 142 times, or 75.13%. Using the past-20-day 20th-percentile dollar volume reduces this to 62 cases, or 32.80%, while mean deployed capital falls to 72.90%. This retrospective risk/utilization trade-off is neither a physical-feedback benefit, an actual execution-failure record, nor an optimal-capacity recommendation.

Positive, conditional, and stopped results all retain an explicit place. The package includes full code, nine capacity CSVs, cash and monotonicity checks, C01–C03 plotting code, and the original certificate.

Appendix E: Research record and reproduction

1. Finding the evidence

The public repository is Econophysics Risk Lab; the research website connects the mathematical model, interactive experiments, findings and applications. The repository README links the shared implementation in `src/econophysics/`, numerical tables in `results/tables/`, study configurations, source records and three executable notebooks. The figure identifiers in this report retain their original F/D/R/P/U/V/T/C/J mapping. The report contains all 19 original scientific figures and eight tables.

TABLE VIII. Reproduction profiles from the repository root. Each run writes new outputs.

Profile	Operation	Evidence
quick	Exact theory, model and source-table checks	Standard-library mathematics and numerical identities
theory	Exact identities and certificates	Calculations under the stated model assumptions
notebooks	Execute all three research notebooks	Model moments, inversion, market tests and forecast comparisons
site	Check local resources and the browser engine	Website links and agreement with Python fixtures
forecast	Download documented archives into a fresh directory and rebuild the fixed experiment	All 183 historical test dates and 16 saved CSV tables
Extended laboratory	Rebuild the four-module HTML from retained source and data	Risk, strategy overlap, allocation and coordination diagnostics

2. Environment and source records

The numerical reconstruction uses Python 3.14.4 and the versions in `requirements-study.txt`; plotting and notebook dependencies are listed separately. The quick checks and website checks also pass on Python 3.11 and 3.14 in the public automation workflows. The three notebooks retain executed outputs. A fresh forecast reconstruction reproduces all 16 saved tables byte for byte. Reproduction entry points and validation records are linked from the README.

Binance Vision-derived results retain CC BY-NC-SA 4.0 attribution and the recorded source terms. Full supplier price histories are obtained through the documented download routes. Code, data and third-party assets retain their respective terms, recorded in `LICENSE.md` and `DATA_LICENSES.md`. Historical experimental digests are preserved; project-name and presentation adaptations are recorded separately from numerical methods and results.